

School of Computer & Systems Sciences

End-Semester Examination, 2023

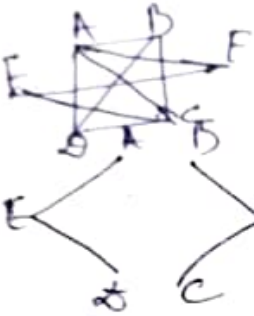
MCA Semester II

Design and Analysis of Algorithms (CS-409)

Time: 2 hours 30 minutes

Maximum Marks: 50

All questions are compulsory.



1/ Define Vertex Cover problem. Show that the Vertex Cover problem is NP-Complete.

2/ Consider working modulo $q = 13$. How many spurious hits does the Rabin-Karp matcher encounter in the text $T = 92654580793$ when looking for the pattern $P = 28$? Show all steps.

3. Using FIFO Branch and Bound, find an optimal solution to the fractional knapsack problem for knapsack weight $(M) = 30$ and, weight values and profit values of four items given by $(10, 10, 12, 18)$ and $(4, 8, 12, 18)$ respectively. Show all steps.

4. Multiply the polynomials $A(x) = 7x^3 - x^2 + x - 10$ and $B(x) = 8x^3 - 6x + 3$ using efficient polynomial-multiplication process. Show all steps.

5. Compute the transitive closure of a directed graph, which is represented by adjacency matrix $\mathbf{W}$

$$\mathbf{W} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Also, computer predecessor matrix. Show all the steps.

[10, 10, 10, 10, 10]